AML Typology Feature Reference

Isolation Forest — TYPOLOGY_FEATURE_PATTERNS

What each feature pattern is capturing, mapped to AML-indicator-DB.xlsx

Overview

The typology-specific Isolation Forest ensemble trains one unsupervised model per AML typology. Each model receives a curated feature set — the TYPOLOGY_FEATURE_PATTERNS — derived directly from the AML-indicator-DB.xlsx indicator catalogue. This document explains what each feature is measuring and which AML indicator it operationalises.

Five typologies are modelled:

- **1.** Structuring & Layering — threshold avoidance and rapid fund velocity
- **2.** Behavioural & Profile Anomalies — activity inconsistent with KYC profile
- **3.** Trade-Based ML & Shell Entities — business volume far exceeding declared activity
- **4.** Cross-Border & Geographic Risk — transactions with high-risk jurisdictions
- **5.** Human Trafficking — MCC-level card and transfer patterns linked to exploitation

1. Structuring & Layering

This typology targets deliberate avoidance of currency-reporting thresholds (STRUCT-001..009) combined with rapid pass-through of funds (ATYPICAL-006..008). The two previously separate typologies — Structuring and Layering — are merged here because both share the same core signals: unusual transaction sizing and high fund velocity.

1.1 Threshold Avoidance

These features detect cash or transaction amounts engineered to stay below regulatory reporting triggers (typically $10,000 CAD).

Feature Name	AML Indicator	What It Captures
`count_near_10k`	STRUCT-003	Number of transactions between $9,000 and $9,999 — the most overt near-threshold band.
`count_very_near_10k`	STRUCT-003	Transactions in a tighter band immediately below $10K, capturing the most deliberate avoidance.

Feature Name	AML Indicator	What It Captures
`count_below_10k`	STRUCT-001	Total count of transactions under $10K, providing volume context for the threshold behaviour.
`ratio_below_10k`	STRUCT-001	Proportion of all transactions that fall below $10K; a high ratio flags systematic splitting, not isolated incidents.
`cash_struct_count`	STRUCT-001	Count of ABM (cash) deposits below $10K. Cash-specific signal because ABM channel identifies physical currency deposits.

1.2 Round-Amount Patterns

Structured payments and layering often involve round-figure transactions that look like controlled fund movement rather than organic commerce.

Feature Name	AML Indicator	What It Captures
`count_round_100`	PROF-007	Number of transactions that are exact multiples of $100. Round amounts in retail contexts are statistically rare and suggest deliberate sizing.
`ratio_round_100`	PROF-007	Share of all transactions that are $100-rounded. A high ratio makes the pattern systemic rather than coincidental.
`count_round_1000`	PROF-007	Number of transactions in exact $1,000 increments. Larger round amounts are a particularly strong structuring signal.
`ratio_round_1000`	PROF-007	Proportion of $1,000-rounded transactions across the observation window.

1.3 Same-Day / Rapid Pass-Through

Funds that arrive and depart the same day suggest the account is being used as a conduit rather than for genuine banking activity.

Feature Name	AML Indicator	What It Captures
`flow_through_ratio`	ATYPICAL-007/008	Ratio of minimum(total inflow, total outflow) to total inflow. A value near 1 means almost everything that comes in goes straight back out — the defining signature of a pass-through account.

1.4 Transaction Velocity

High frequency of transactions in a short period is the core STRUCT-006 signal — layering generates many small movements to obscure the audit trail.

Feature Name	AML Indicator	What It Captures
`transactions_per_active_day`	STRUCT-006	Average daily transaction count on days the account is active. High values indicate burst activity inconsistent with normal account use.
`volume_per_day`	STRUCT-006	Total transaction volume divided by the observation window in days. Captures sustained high-throughput even when individual transactions are small.

1.5 Location Hopping

Using ATMs or transaction points across multiple cities and provinces is a BEHAV-001 signal, indicating deliberate geographic dispersal to avoid pattern detection at a single branch.

Feature Name	AML Indicator	What It Captures
`unique_cities`	BEHAV-001	Number of distinct cities where the customer transacted. Multi-city ABM use is a primary human trafficking and structuring signal.
`unique_provinces`	BEHAV-001	Number of distinct provinces spanned. Cross-provincial activity raises the geographic risk further.

1.6 Channel Volume — Rapid Fund Movement

The total counts and sums by payment channel provide the raw throughput context for structuring and layering. Large or frequent EFT/EMT/wire transactions alongside ABM cash point to rapid, multi-channel fund movement (ATYPICAL-007, PROF-006).

Feature Name	AML Indicator	What It Captures
`count_abm / sum_abm`	STRUCT-001	ABM transaction count and total value. High ABM volume anchors the cash-structuring signal.
`count_eft / sum_eft`	ATYPICAL-007	EFT count and value. Immediate EFT outflows following cash deposits flag rapid pass-through layering.
`count_emt / sum_emt`	ATYPICAL-007	EMT (e-transfer) count and value. Fast person-to-person transfers are a low-oversight layering channel.
`count_wire / sum_wire`	PROF-006	Wire count and total. Large wires move money rapidly and across borders, consistent with large-scale layering.
`count_cheque`	STRUCT-009	Cheque volume. Cheques can be used to layer funds while avoiding electronic audit trails.

2. Behavioural & Profile Anomalies

This typology captures activity that is inconsistent with what would be expected given a customer's declared KYC profile — occupation, income, account history, and prior behaviour. It consolidates

the former money-mule and profile-mismatch typologies, since both target PROF and ACCT indicators.

2.1 Spending vs Income Mismatch

Living beyond one's means is a defining AML indicator (PROF-002, PROF-005). These features measure the gap between what flows out and what the customer's profile would predict as income.

Feature Name	AML Indicator	What It Captures
`spending_to_income_ratio`	PROF-002/005	Total debits divided by estimated or declared income. Values significantly above 1.0 indicate spending that cannot be explained by legitimate income.
`income_vol_ratio`	PROF-001	Transaction volume relative to expected account activity established at onboarding. Captures accounts that dramatically exceed their projected usage.

2.2 Sudden Changes in Activity

Abrupt shifts in transaction patterns — not just high volume — are the key PROF-010 and ACCT-004 signals. A previously quiet account that suddenly processes large flows is anomalous even if the absolute values are not extreme.

Feature Name	AML Indicator	What It Captures
`monthly_volume_cv`	PROF-010/ACCT-004	Coefficient of variation of monthly transaction volume. High CV means the customer has months of very different activity levels — sudden spikes or drops.
`monthly_txn_count_std`	ACCT-002	Standard deviation of the monthly transaction count. Complements volume CV by capturing erratic transaction frequency independent of amount.
`amount_cv`	PROF-008	Coefficient of variation of individual transaction amounts. High CV means transaction sizes are highly irregular, inconsistent with a predictable income-and-spend pattern.
`std_transaction_amount`	PROF-008	Standard deviation of transaction amounts in absolute terms. Provides scale-sensitive context alongside the ratio-based CV.

2.3 Transaction Size Context

The distribution of transaction amounts — maximum, average — helps profile whether a customer's transactions are typical for their stated financial situation.

Feature Name	AML Indicator	What It Captures
`max_transaction_amount`	PROF-008	Largest single transaction. Unusually large one-off transactions relative to the customer's normal activity are a profile mismatch signal.

Feature Name	AML Indicator	What It Captures
`avg_transaction_amount`	PROF-008	Mean transaction size. Very high or very low averages relative to peer cohorts flag atypical transaction sizing.

2.4 Channel Use Patterns

Money mules and profile mismatches often use an unusual mix of payment channels, either highly concentrated in one channel or spanning all channels in ways inconsistent with their declared purpose.

Feature Name	AML Indicator	What It Captures
`channel_diversity`	ACCT-002	Count of distinct payment channels used (ABM, EFT, EMT, wire, card, cheque). Very low or very high diversity can both be anomalous depending on the customer type.
`channel_concentration_hhi`	ACCT-002	Herfindahl-Hirschman Index of channel use. A value near 1 means the customer is overwhelmingly concentrated in one channel — unusual for most legitimate customers.
`count_card / sum_card`	ACCT-003	Card transaction count and total. A sudden surge in card usage is the ACCT-003 indicator — particularly relevant for accounts that previously transacted primarily via other channels.

2.5 Overall Volume and Activity Window

Raw throughput and account tenure contextualise whether the transaction volume is plausible for the customer's profile.

Feature Name	AML Indicator	What It Captures
`transaction_count_total`	PROF-003	Total number of transactions in the observation window. Compared against geographic and demographic norms to detect outliers.
`net_flow`	PROF-001	Total inflow minus total outflow. A near-zero net flow on a high-volume account is a strong pass-through or mule signal.
`active_days`	ACCT-004	Number of days with at least one transaction. Low active days relative to time span indicates burst activity — not continuous organic use.
`time_span_days`	ACCT-001	Observation window length. Provides the denominator for velocity calculations and tenure context.
`customer_tenure_days`	ACCT-001	Days since account opening. New accounts with high activity are disproportionately associated with money-mule and fraud typologies.

2.6 Customer Demographics

Demographic attributes help calibrate whether behaviour is expected for this type of customer and flag known high-risk profile combinations.

Feature Name	AML Indicator	What It Captures
`is_individual`	PROF-002	Binary flag: personal vs business account. Individuals and businesses have different baseline behaviours; the model learns different anomaly thresholds for each.
`age`	PROF-002	Customer age. Very young or elderly customers with large, complex transactions are disproportionately targeted for exploitation.
`occupation_risk_high`	PROF-002	Flag for occupations with elevated AML exposure (e.g. cash-intensive businesses, politically exposed roles). Amplifies other signals when present.

3. Trade-Based ML & Shell Entities

This typology targets businesses whose financial flows are inconsistent with their declared commercial activity (PML-TBML-02..08, PROF-004). It also captures gatekeeper pass-through via GATE-001 — accounts used by professionals (lawyers, accountants, realtors) as conduits for client funds.

3.1 Volume vs Declared Business Activity

The foundational trade-based ML signal: a business transacts far more than its declared sales revenue or industry norms would suggest.

Feature Name	AML Indicator	What It Captures
`volume_to_sales_ratio`	PROF-004/PML-TBML-05	Total transaction volume divided by the customer's declared annual sales. A ratio well above 1 means the business is moving more money than it claims to earn — the core TBML signal.
`inflow_per_day`	PML-TBML-04	Average daily inflow. Sudden large inflows that are inconsistent with the business's normal operating rhythm flag the PML-TBML-04 sudden-spike indicator.
`total_inflow`	PML-TBML-04	Cumulative inflow over the observation window. High totals relative to declared business size confirm the volume mismatch.

3.2 Industry and Entity Risk

Certain industries and entity structures carry elevated inherent TBML risk regardless of transaction patterns.

Feature Name	AML Indicator	What It Captures
`industry_risk_high`	PML-TBML-02	Flag for high-risk industry codes (e.g. import/export, precious metals, real estate, currency exchange). Sector deviation — a business whose transactions don't match its stated sector — is PML-TBML-02.
`is_business`	PROF-004	Binary flag: business account. All trade-shell signals are amplified for business customers; individual customers are downweighted in this typology.
`employee_count`	Shell signal	Number of employees on record. High transaction volumes combined with very few employees is a classic shell company indicator — maximum revenue, minimal operating footprint.

3.3 Pass-Through / Gatekeeper

Gatekeepers — lawyers, notaries, accountants — legitimately hold client funds, but an account used purely as a conduit with no economic retention is anomalous (GATE-001).

Feature Name	AML Indicator	What It Captures
`flow_through_ratio`	GATE-001	As in structuring, this ratio near 1 means funds enter and exit without being retained. For business accounts this is the primary gatekeeper signal.

3.4 EFT Inflow Spikes

PML-TBML-04 specifically calls out sudden large EFT inflows. These are captured from the raw EFT transaction file rather than derived features, giving the model access to the direct signal.

Feature Name	AML Indicator	What It Captures
`eft_credit_sum`	PML-TBML-04	Total EFT credit (inflow) received. High values relative to declared sales are the clearest TBML-04 signal.
`eft_credit_count`	PML-TBML-04	Number of EFT credit transactions. Frequent large inflows are more anomalous than a single payment.
`sum_eft / count_eft`	PML-TBML-04	Broader EFT totals across debit and credit, providing context for the credit-specific features above.

3.5 Wire Activity

High-value wires to multiple counterparties (PML-TBML-03) indicate a business distributing funds widely — potentially integrating layered proceeds.

Feature Name	AML Indicator	What It Captures
`sum_wire`	PML-TBML-03	Total wire transfer value. Large wire totals relative to declared business income flag illicit integration.

Feature Name	AML Indicator	What It Captures
`count_wire`	PML-TBML-03	Number of wire transfers. Many wires to different parties operationalises the 'large number of entities' signal in PML-TBML-03.
`has_wire`	PML-TBML-03	Binary flag: does the customer use wires at all? For low-revenue businesses, any wire activity is anomalous.
`has_eft`	PML-TBML-04	Binary flag: does the customer use EFT? Complements has_wire for channel-presence detection.

3.6 Round-Amount Invoices

PML-TBML-08 highlights round-figure payments typical of fictitious invoicing — a common mechanism to move funds under the cover of trade documents.

Feature Name	AML Indicator	What It Captures
`ratio_round_1000`	PML-TBML-08	Proportion of transactions in exact $1,000 increments. For business accounts, this proxies for round-figure invoice payments (the indicator references ~$50K increments but the pattern manifests across round amounts generally).
`count_round_1000`	PML-TBML-08	Absolute count of $1,000-rounded transactions.

3.7 Spike Pattern Detection

Beyond the total volume, the temporal distribution of transactions reveals whether inflows are consistently high or arrive as sudden, unexplained spikes.

Feature Name	AML Indicator	What It Captures
`monthly_txn_cv`	PML-TBML-04	Coefficient of variation of monthly transaction counts. A single month with dramatically elevated activity (high CV) is the spike signature of PML-TBML-04.

4. Cross-Border & Geographic Risk

This typology captures transactions with jurisdictions that carry elevated money laundering or terrorist financing risk (GEO-001..005) and wire transfer patterns inconsistent with the customer's financial profile (WIRE-008/010).

4.1 High-Risk Jurisdiction Exposure

Each feature below maps to a specific geographic risk category from the AML indicator database.

Feature Name	AML Indicator	What It Captures
`drug_country_txn_count`	GEO-001	Transaction count with drug-producing or drug-transit countries (e.g. Mexico, Colombia, Peru, Bolivia). Direct operationalisation of the narcotics-source jurisdiction indicator.
`high_risk_fatf_txn_count`	GEO-002/004	Transactions with FATF blacklisted jurisdictions (North Korea, Iran, Myanmar). The most severe geographic risk tier.
`greylist_txn_count`	GEO-003	Transactions with FATF greylisted countries — jurisdictions under increased monitoring. A softer but broader risk signal than the blacklist.
`offshore_center_txn_count`	GEO-003	Transactions with known offshore financial centres (e.g. Cayman Islands, BVI, Panama). These jurisdictions are associated with secretive banking and layering.
`underground_banking_country_count`	GEO-003	Count of jurisdictions known for underground banking networks (e.g. China, Hong Kong, Philippines). Captures the hawala and fei-ch'ien risk.

4.2 International Activity Volume

Beyond specific high-risk countries, the overall volume and frequency of cross-border transactions relative to the customer's profile is the GEO-005 signal.

Feature Name	AML Indicator	What It Captures
`international_txn_count`	GEO-005	Total number of transactions to or from foreign counterparties. Frequency of overseas transfers is the primary GEO-005 indicator.
`international_txn_sum`	GEO-005	Total value of international transactions. Large cross-border volumes inconsistent with income or business size are anomalous.
`international_ratio`	GEO-005	Proportion of all transactions that are cross-border. A high ratio means the customer's financial activity is predominantly international — unusual unless the profile supports it.
`unique_countries`	GEO-005	Number of distinct countries transacted with. Wide geographic scatter is harder to explain through legitimate activity than a few recurring partners.
`has_any_high_risk_txn`	GEO-002/004	Binary flag: at least one transaction with any high-risk jurisdiction. Ensures even single-transaction exposure is captured when volume-based features may be small.

4.3 Western Union / Money Service Businesses

Western Union and similar MSBs are inherently international and carry elevated risk due to limited AML oversight in many corridors.

Feature Name	AML Indicator	What It Captures
`sum_westernunion`	GEO-005	Total value sent via Western Union or equivalent MSBs. High values, especially to high-risk corridors, directly operationalise GEO-005.
`count_westernunion`	GEO-005	Number of Western Union transactions. Frequent MSB use is anomalous for most retail banking profiles.
`has_western_union`	GEO-005	Binary flag: any WU activity. Presence alone is an elevated-risk signal for customers with no declared cross-border activity in their KYC.

4.4 Wire Transfer Patterns

WIRE-008 and WIRE-010 target large-volume foreign wire transfers and wires involving multiple counterparties — both consistent with layering through cross-border channels.

Feature Name	AML Indicator	What It Captures
`sum_wire`	WIRE-008	Total wire value. Large foreign wires relative to the customer's income or declared business are the WIRE-008 signal.
`count_wire`	WIRE-010	Wire transaction count. Multiple wires to different parties is the WIRE-010 indicator.
`card_unique_countries`	GEO-005	Number of countries where the customer's card has been used at merchants. Card data now provides a direct geographic footprint beyond wire and EFT channels.

5. Human Trafficking

This typology operationalises the full HT-SEX-01..14 indicator set. Rather than relying on generic transaction aggregates, the features are built directly from Merchant Category Codes (MCCs) on card transactions and from the EFT/EMT transaction files, enabling precise detection of the spending patterns associated with sexual exploitation.

5.1 After-Hours Spa, Massage & Escort Spend (HT-SEX-12)

Card transactions at spa, massage, and personal care establishments during late-night or early-morning hours are the strongest single indicator of commercial sexual exploitation revenue.

Feature Name	AML Indicator	What It Captures
`card_adult_afterhours_count`	HT-SEX-12	Number of card transactions at adult entertainment, spa, or massage MCCs (7297,

Feature Name	AML Indicator	What It Captures
		7298) outside normal business hours. This is the most specific HT card signal and is sourced directly from the card transaction file.
`card_afterhours_count`	HT-SEX-12	Broader count of all card transactions occurring after hours, regardless of merchant category. Provides context for whether after-hours activity is general or concentrated in the adult-sector MCCs.
`night_transaction_ratio`	HT-SEX-12	Proportion of all transactions (any channel) occurring at night. A high ratio across channels, not just card, reinforces the after-hours lifestyle pattern.
`weekend_ratio`	HT-SEX-12	Proportion of transactions occurring on weekends. Trafficking operations tend to concentrate activity on weekends; a high ratio amplifies the after-hours signal.

5.2 Multi-City ABM Cash Withdrawals (HT-SEX-07)

Traffickers controlling victims across multiple locations need cash in each city. Multi-city ABM withdrawals at night are the clearest operational signal of this movement pattern.

Feature Name	AML Indicator	What It Captures
`night_abm_count`	HT-SEX-07	Count of ABM withdrawals occurring at night. Cash extracted late at night is the primary operational currency for trafficking.
`unique_cities`	HT-SEX-07/08	Number of distinct cities where the customer transacted. Shared with the structuring typology but here interpreted as multi-city victim management or client servicing.
`unique_provinces`	HT-SEX-07/08	Number of provinces spanned. Cross-provincial movement indicates inter-city trafficking routes.

5.3 Accommodation Spend (HT-SEX-08)

Hotels and short-term accommodation in cities other than the customer's residential city are used as working venues. Card spend at accommodation MCCs in non-residential locations is a direct proxy.

Feature Name	AML Indicator	What It Captures
`card_accommodation_count`	HT-SEX-08	Number of card transactions at hotel and accommodation MCCs (7011). High counts in combination with multi-city presence confirm the non-residential venue pattern.
`card_accommodation_sum`	HT-SEX-08	Total spend on accommodation. Repeated hotel stays represent both operational cost and a strong geographic displacement signal.

Feature Name	AML Indicator	What It Captures
`ht_accommodation_sector`	HT-SEX-07	Binary flag: is the customer's declared business in the accommodation or hospitality sector? Front businesses in this sector are commonly used to co-mingle trafficking proceeds.

5.4 Luxury Spend vs Income (HT-SEX-03)

Traffickers often display conspicuous luxury consumption — jewellery, high-end goods — that is inconsistent with declared income. Card MCCs for luxury retail combined with the income-to-spending ratio surface this pattern.

Feature Name	AML Indicator	What It Captures
`card_luxury_sum`	HT-SEX-03	Total card spend at luxury MCCs (e.g. jewellers 5094, 5944). High luxury spend relative to income is a defining HT-SEX-03 signal.
`card_luxury_count`	HT-SEX-03	Number of luxury merchant transactions. Frequency distinguishes lifestyle spending from isolated purchases.
`spending_to_income_ratio`	HT-SEX-13	Shared with the behavioural typology; here it proxies HT-SEX-13 — high-value transfers disproportionate to declared income, indicating unexplained wealth from trafficking proceeds.

5.5 Travel to Source Countries (HT-SEX-10)

Airfare and travel spend to known human trafficking source countries — those with high rates of trafficking outflows — indicates potential recruitment or victim transport.

Feature Name	AML Indicator	What It Captures
`card_travel_sum`	HT-SEX-10	Total card spend on travel MCCs (airlines 4511, travel agencies 4722). Large travel spend, especially to high-risk source countries, is the HT-SEX-10 signal.
`card_travel_count`	HT-SEX-10	Number of travel transactions. Frequent trips signal recurring movement, not one-time tourism.
`ht_source_country_txn_count`	HT-SEX-10	Transactions (any channel) with known trafficking source countries. Built from a curated high-risk origin country list and sourced directly from the transaction files.
`sum_westernunion`	HT-SEX-10	WU transfers to source countries are a common mechanism for paying recruiters or remitting proceeds to home jurisdictions.

5.6 Victim Maintenance Spending (HT-SEX-04/05)

Traffickers cover victims' day-to-day needs — food, parking, transport — as a means of maintaining control. These mundane spending categories become anomalous when concentrated, frequent, and paired with the other HT signals.

Feature Name	AML Indicator	What It Captures
`card_parking_count`	HT-SEX-04	Number of parking transactions (MCC 7523). Frequent parking charges across multiple cities indicate victim transit management rather than personal commuting.
`card_delivery_count`	HT-SEX-05	Number of food delivery transactions (MCC 5814 and delivery apps). Regular food delivery to accommodation venues, especially after hours, is consistent with victim maintenance.

5.7 Digital, Crypto & Gambling (HT-SEX-06)

Virtual currency purchases and online gambling are used to launder trafficking proceeds and pay for digital advertising of services.

Feature Name	AML Indicator	What It Captures
`card_digital_count`	HT-SEX-06	Card transactions at digital goods, virtual currency, and online gambling MCCs (4816, 7995). These channels offer anonymity for both proceeds laundering and operational payments.

5.8 Retail & Gift Card Purchases (HT-SEX-01/02)

Gift cards and retail purchases (grocery, pharmacy) serve as untraceable payment instruments for both victim management and advertising on illicit platforms.

Feature Name	AML Indicator	What It Captures
`card_retail_count`	HT-SEX-01/02	Card transactions at retail and convenience MCCs (5411 grocery, 5912 pharmacy, 5310 discount). High volumes suggest bulk gift card or supply purchases associated with trafficking operations.

5.9 Rental Payments via EMT / EFT (HT-SEX-14)

Rent paid by e-transfer or EFT for apartments used as venues is a direct financial footprint of the trafficking operation. These payments are captured from the raw EMT and EFT files rather than from card data, ensuring no channel is missed.

Feature Name	AML Indicator	What It Captures
`emt_debit_count`	HT-SEX-14	Number of EMT (e-transfer) debit outflows. Recurring EMT debits of consistent amounts correspond to rental and operational payments for trafficking venues.

Feature Name	AML Indicator	What It Captures
`eft_debit_count`	HT-SEX-14	Number of EFT debit outflows. EFT-based rental payments complement EMT and ensure coverage regardless of how the venue payment is structured.